

Deep Learning driven automated person detection and tracking model on surveillance videos

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ABSTRACT

Recently, person detection and tracking in a video scene of a surveillance system were grabbing higher interest because of its extensive range of applications in gender classification, abnormal event recognition, person identification, human gait characterization, individual counting in a dense crowd, and fall detection for older people, etc. Several methodologies were implied for person detection for surveillance applications. This paper presents a new DL driven automated person detection and tracking (DLD-APDT) model on surveillance videos. The major goal of the proposed DLD-APDT model is to identify people and track them in the videos. To accomplish this, the proposed DLD-APDT model initially performs frame conversion process where the input video is converted into a set of frames. In addition, the proposed DLD-APDT model employs EfficientDet model as object detector, i.e., persons and track them. EfficientDet is a recently developed object detector that makes use of different optimization and backbone tweaks, like bi-directional feature pyramid network (BiFPN) and a compound scaling approach. Moreover, root means square propagation (RMSProp) optimizer is applied to optimally choose the hyperparameters related to the EfficientNet model which helps in accomplishing enhanced performance. The experimental validation of the DLD-APDT model takes place using two training datasets namely PascalVOC and PenFudan datasets. The simulation results reported the enhanced detection and tracking performance of the DLD-APDT technique over recent approaches.

1. Introduction

Person recognition is regarded as an appropriate subset of the generic image related detection task with several probable application zones like human-robot interaction, security, or surveillance, currently the social security in an epidemic context. Person recognition is a difficult task which was broadly discovered for side-mounted, rectilinear-lens cameras [1,2]. Implementation of such systems to overhead fisheye (OF) cameras is not unimportant because of the severe geometric distortions and radial geometry in the captured images. In spite of important development in the past 2 years [3], fisheye person recognition turns out to be a difficult and comparatively unrevealed issue, mainly owing to the restricted diversity of datasets for evaluating and training purposes [4]. Object detection and tracking have acquired rising interest owing to their extensive array of applications and current breakthrough research. In the applications of academia and real-world

object detection and tracking contains equal significance. Certain real-world applications involve transportation surveillance, autonomous driving, robotic vision, and monitoring security [5].

The boom of Artificial Intelligence (AI) has entered every walk of life in community, and research on AI and Deep Learning (DL) has done breakthroughs [6]. In DL structure, the issue of object detection is observed as a mission of labeling distinct objects in an image frame with their right classes and forecasting its bounding boxes with a high possibility. The learning outcome in DL is reliable on the quantity of specimens (or earlier experiences). If the number is larger, then the outcome is very precise [7]. Nowadays, plentiful data that form DL is a useful choice. But DL frequently requires thousands or hundreds of images for acquiring the superior outcomes, dissimilar to the classical (shallow) learning. The research which is ongoing on multi-camera person detection includes hand-crafted features, however to the best of our knowledge real-world industrial applications utilize monocular CNN

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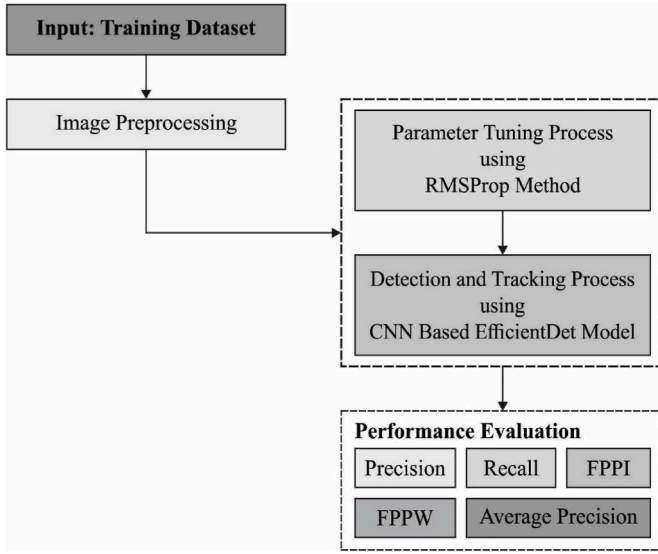


Fig. 1. Overall process of DLD-APDT algorithm.

pedestrian detectors, and the final valuation can be performed by averaging the separate per view ones [8,9].

Al Atrash [10] established 2 methods to detect and count people's faces. These techniques were dependent upon Faster-RCNN and SSD that are approaches utilized to object recognition. It can utilize these techniques for Face detection and calculating these faces. These approaches were DNNs that are trained on object recognition tasks. During this case, it can be trained Faster-RCNN and SSD techniques on Wider-Face dataset that is collected of faces from a variety of criteria compared to expression, pose, occlusion, illumination, and scale. In Ref. [11], the authors projected a Kalman filter (KF) based on CNN to online pedestrian recognition from the video. Primary, the single shot multi-box detector was executed as CNN detector that integrates the pedestrian aspect ratios. The fusion elements were executed for improving the detector robustness to medium and far scale pedestrians. Next, the bounding box propagates based on the forecast in KF. At last, the place and confidence of bounding boxes were refined by KF.

Wang and Lan [12] presented a new infrared pedestrian recognition network from a per-pixel forecast manner. The presented network has 3 parts containing neck, head, and backbone: i) The backbone increases the receptive domain of networks and resolves the network degradation issue produced by network deepening; ii) The neck enhances feature illustration by fusing multiscale feature data; iii) With planning several

cascaded heads, pixel-by-pixel forecast of multiscale pedestrian was attained. The authors in Ref. [13] developed a novel hybrid 2D and 3D convolutional related RNN (HCRN) to video-related person re-id task. Especially, the 3D convolution element explores the local short-term fast varying motion data, but the recurrent layer leverages the global long-term spatial-temporal data. According to HCRN, it can be planned a scale-fusion structure for making entire utilize of features of the distinct scales to enhance the efficiency of video based person re-id. In Ref. [14], the authors attempt for improving classifier procedure with several approaches. A particularly adapted CNN framework has been utilized together with LSTM and time distribution from experimental. The convolutional neural layer for 2D data from the structure is utilized.

This paper presents a new DL driven automated person detection and tracking (DLD-APDT) model on surveillance videos. The proposed DLD-APDT model initially performs frame conversion process where the input video is converted into a set of frames. In addition, the proposed DLD-APDT model employs EfficientDet model as object detector, i.e. persons and track them. EfficientDet is a recently developed object detector that makes use of different optimization and backbone tweaks, like bi-directional feature pyramid network (BiFPN) and a compound scaling approach. Moreover, root means square propagation (RMSProp) optimizer is applied to optimally choose the hyperparameters. The experimental validation of the DLD-APDT model takes place using two training datasets namely PascalVOC and PenFudan datasets.

2. The proposed model

In this paper, a new DLD-APDT model was enhanced for effectual person detection and tracking on surveillance videos. The proposed DLD-APDT model initially performs frame conversion process where the input video is converted into a set of frames. Followed by, the proposed DLD-APDT model utilized EfficientDet model to detect persons and track them. Finally, the RMSProp optimizer is applied to optimally choose the hyperparameters. Fig. 1 demonstrates the overall process of DLD-APDT algorithm.

2.1. Object detection and tracking: EfficientDet model

In this study, the presented DLD-APDT model utilized EfficientDet model to detect persons and track them. A new family of detection paradigms called EfficientDet is established based on the BiFPN. The complete structure of EfficientDet mainly follows the one-stage detector models [15]. Then, we use ImageNet-pre-trained EfficientNet as the backbone network (BBN). The presented model acts as a featured network that takes level1 3–7 features $\{P_3, P_4, P_5, P_6, P_7\}$ from the BBN and repetitively employs the combination of top-down and bottom-up

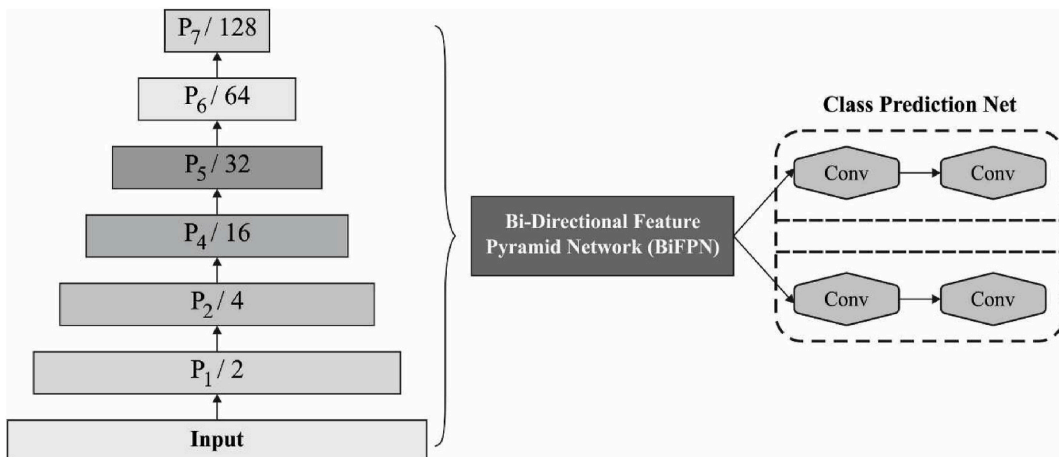


Fig. 2. EfficientDet architecture.

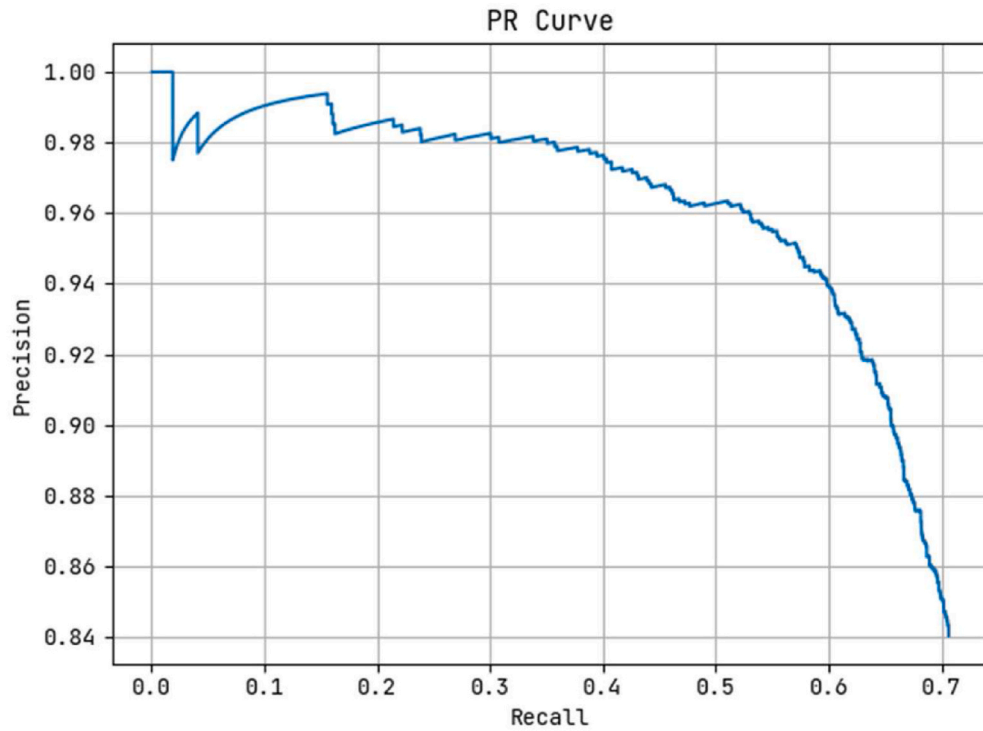


Fig. 3. Precision-recall review of DLD-APDT algorithm in PascalVOC dataset.

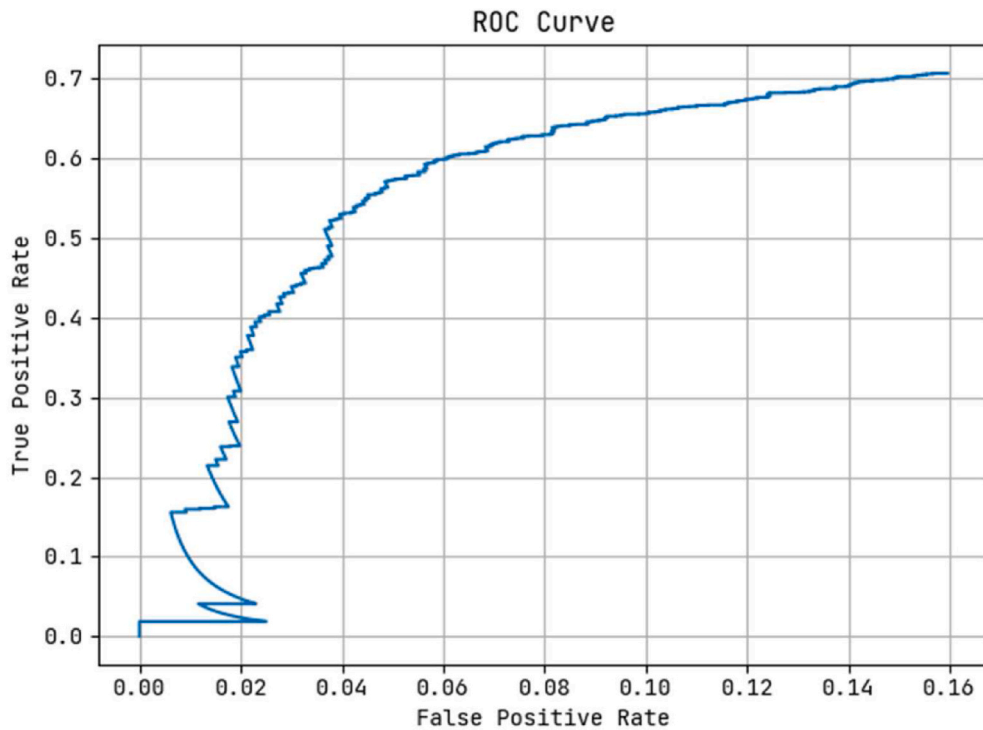


Fig. 4. ROC analysis of DLD-APDT technique under PascalVOC dataset.

bi-directional features.

Aiming at enhancing efficiency and accuracy, we want to improve a family of model which could meet a wider range of restraints. Here, a crucial problem is scaling up a benchmark EfficientDet mechanism. Fig. 2 depicts the framework of EfficientDet approach. Earlier studies mainly scale up a benchmark detector by using larger BBNs (for example, AmoebaNet or ResNeXt), stacking more FPN layers, or using

larger input images. Usually, this method is ineffectual because it emphasizes only a limited or single scaling dimension. The current study exhibits outstanding performance in image categorization by equally scaling up each dimension of network input, width, and depth resolution. Based on the study, we present a new compound scaling technique for detecting objects that make use of a ϕ compound coefficient to equally scale up each dimension of resolution, backbone, BiFPN, and

Table 1

Overall analysis of DLD-APDT approach with various measures under PascalVOC dataset.

Measures	Values
Precision	92.95
Recall	61.86
FPPI	0.047
FPW	0.071
Average Precision	69.94

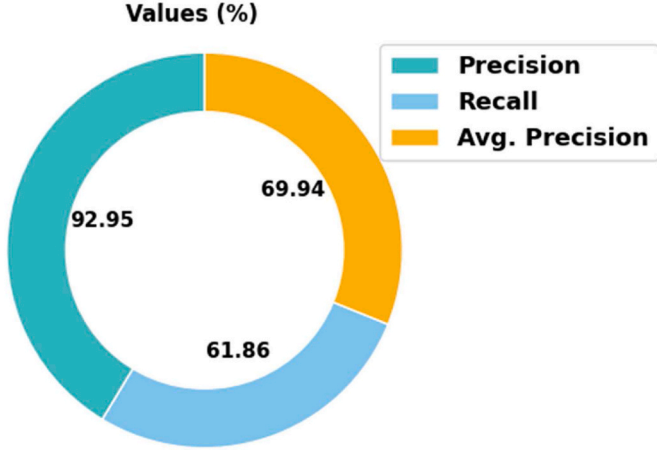


Fig. 5. Overall analysis of DLD-APDT approach under PascalVOC dataset.

class/box network. In contrast, object detector has considerably larger scaling dimension when compared to image classification model, hence grid search for each dimension is cost-consuming. Thus, we employ a heuristic-based scaling model, however following the major concept of equally scaling up each dimension.

BBN — we reutilize the same depth or width scaling coefficient of EfficientNet-B0 to B6 so we simply reutilize the ImageNet-pretrained

checkpoint.

BiFPN network — we rise BiFPN depth D_{bifpn} (#layers) linearly because depth should be rounded to smaller integers. For BiFPN width W_{bifpn} (#channel), rapidly raise BiFPN width W_{bifpn} (#channel). Select the optimal value 1.35 as the BiFPN width scaling factor. In the following, BiFPN width and depth were scaled by using the below expression:

$$W_{bifpn} = 64 \cdot (1.35^\varphi), D_{bifpn} = 3 + \varphi \quad (1)$$

Box/class predictive networks— it can be fixing the width to be often similar to the BiFPN (viz., $W_{pred} = W_{bifpn}$), however, increase the depth (#layers) linearly using the subsequent formula:

$$D_{box} = D_{class} = 3 + \lfloor \varphi/3 \rfloor \quad (2)$$

Input image resolution- features level3-7 are utilized in BiFPN, the input resolution should be divisible by $2^7 = 128$, hence we rise resolution linearly by the following expression:

$$R_{input} = 512 + \varphi \cdot 128 \quad (3)$$

Following Eqs. (1)–(3) with diverse φ , it is designed EfficientDet-D0 ($\varphi = 0$) to D7 ($\varphi = 7$), where D7 and D7x have a similar head and BiFPN, but D7 employs high resolution and D7x employs large BBN and additional feature levels (from P_3 to P_8). Especially, the compound scaling is heuristic-based and mightn't be optimum, but shows that the scaling model could considerably enhance efficacy when compared to other single-dimension scaling models.

2.2. Hyperparameter tuning: RMSProp optimizer

In this work, the RMSProp optimizer is applied to optimally choose the hyperparameters related to the EfficientNet model which helps in accomplishing enhanced performance. The RMSprop optimizer is similar to the gradient descent (GD) technique with momentum [16]. The RMSprop optimizing restricts the oscillation from the upright way. Thus, it improves our rate of learning and this technique is take substantial steps from the horizontal way and converges rapidly. The variance amongst RMSprop and GD is on how the gradient is computed.

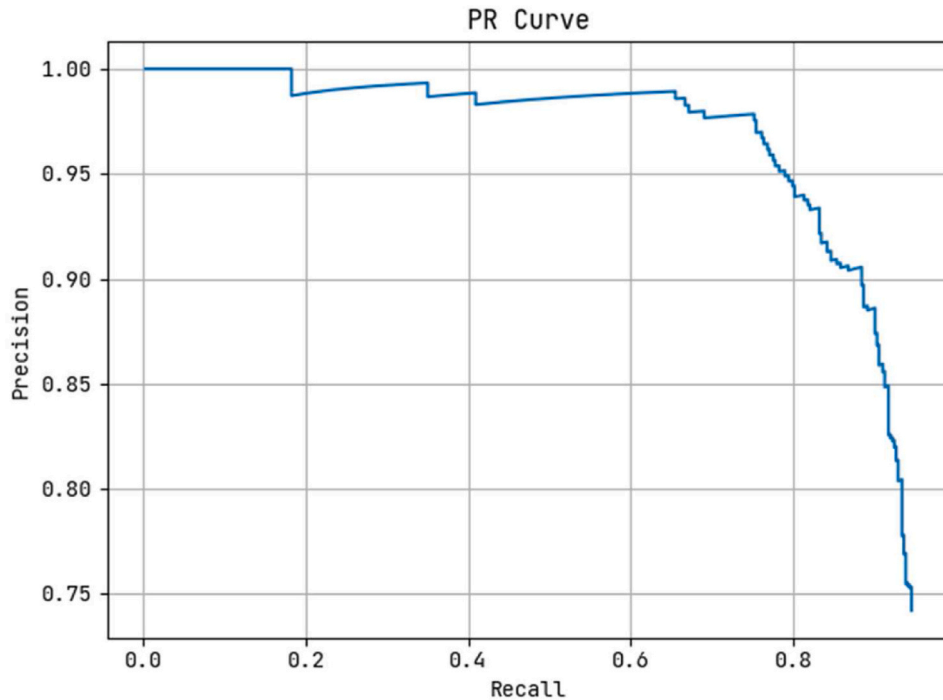


Fig. 6. Precision-recall interpretation of DLD-APDT algorithm under PenFudan dataset.

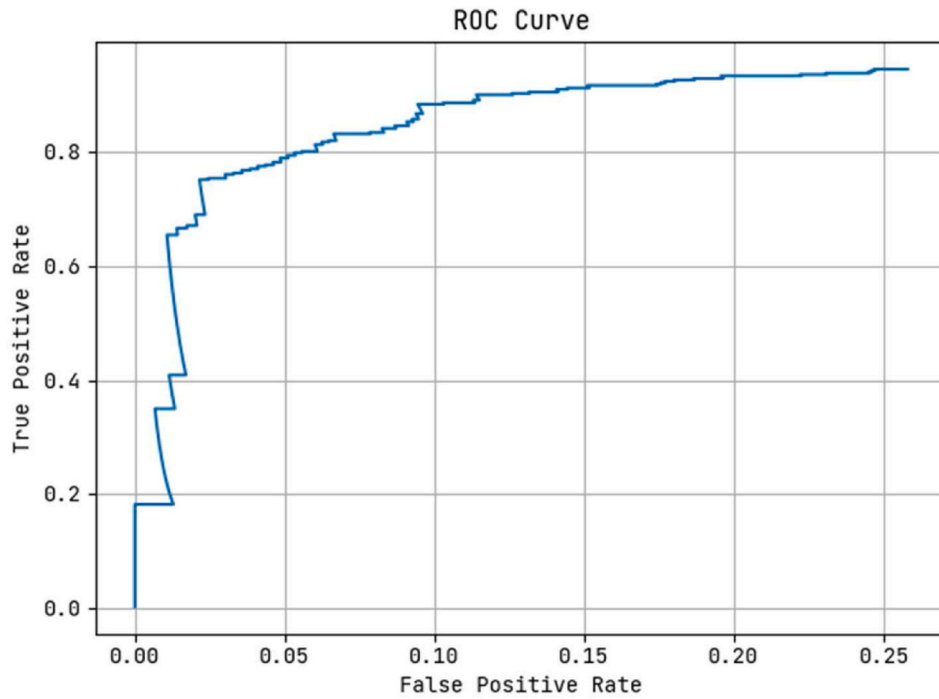


Fig. 7. ROC analysis of DLD-APDT technique under PenFudan dataset.

Table 2

Overall analysis of DLD-APDT approach with various measures under PenFudan dataset.

Measures	Values
Precision	86.46
Recall	90.54
FPPI	0.141
FPPW	0.134
Average Precision	88.76

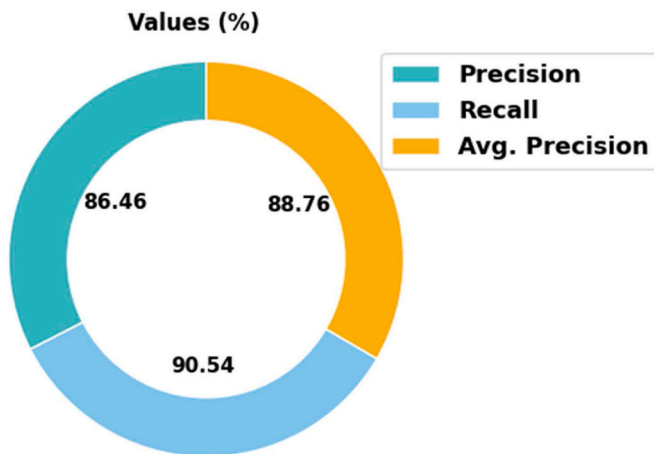


Fig. 8. Overall analysis of DLD-APDT approach under PenFudan dataset.

It can be computing the running average with respect to means square as demonstrated in Eq. (4):

$$v(w, t) = \gamma v(w, t-1) + (1-\gamma)(\nabla Q_i(w))^2 \quad (4)$$

In Eq. (4), γ refers the forgetting factor

$$w := w - \frac{n}{\sqrt{v(w, t)}} \nabla Q_i(w) \quad (5)$$

Eq. (5) depicts the upgrade of parameters.

3. Results and discussion

The experimental validation of the DLD-APDT model is trained using two datasets namely Pascal VOC [17] and PenFudan [18] dataset. Fig. 3 provides a clear precision-recall curve of the DLD-APDT model attained under PascalVOC dataset. The figure implied that the DLD-APDT model has shown effectual outcome on the applied datasets.

The person detection and tracking performance of the DLD-APDT model obtained under the PascalVOC dataset are illustrated in Fig. 4. The figure highlighted that the DLD-APDT method has shown an effectual outcome with maximum ROC values.

Table 1 and Fig. 5 provide an overall person detection performance of the DLD-APDT model on the applied PascalVOC dataset. The results indicated that the DLD-APDT method has obtained improved results with the maximum $prec_n$, $recal$, and average $prec_n$ of 92.95%, 61.86%, and 69.94%. It also attained minimal False Positive Per Image (FPPI) and False Positive Per Window (FPPW) values of 0.047 and 0.071 respectively.

Fig. 6 offers a clear precision-recall curve of the DLD-APDT approach reached in PenFudan dataset. The figure shows the DLD-APDT algorithm has revealed effective outcome on the applied datasets.

The person detection and tracking performance of the DLD-APDT methodology gained under the PenFudan dataset are showcased in Fig. 7. The figure examined that the DLD-APDT method has outperformed effectual outcomes with maximal ROC values.

Table 2 and Fig. 8 offer an overall person detection performance of the DLD-APDT algorithm on the applied PenFudan dataset. The outcomes referred that the DLD-APDT system has gained higher outcomes with the maximal $prec_n$, $recal$, and average $prec_n$ of 86.46%, 90.54%, and 88.76%. It also achieved minimal FPPI and FPPW values of 0.141 and 0.134 correspondingly.

During the testing process, the result analysis of the DLD-APDT model is tested using two videos namely video 1 [19] and video 2

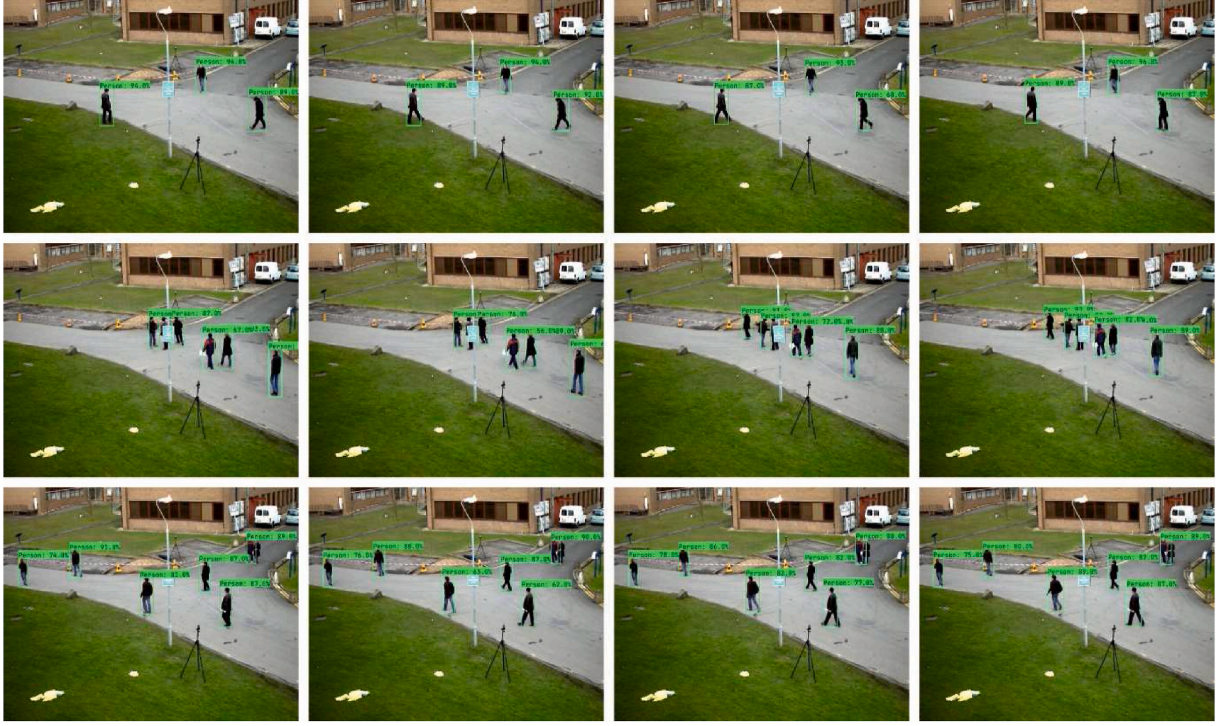


Fig. 9. Sample person detection on test video 1.

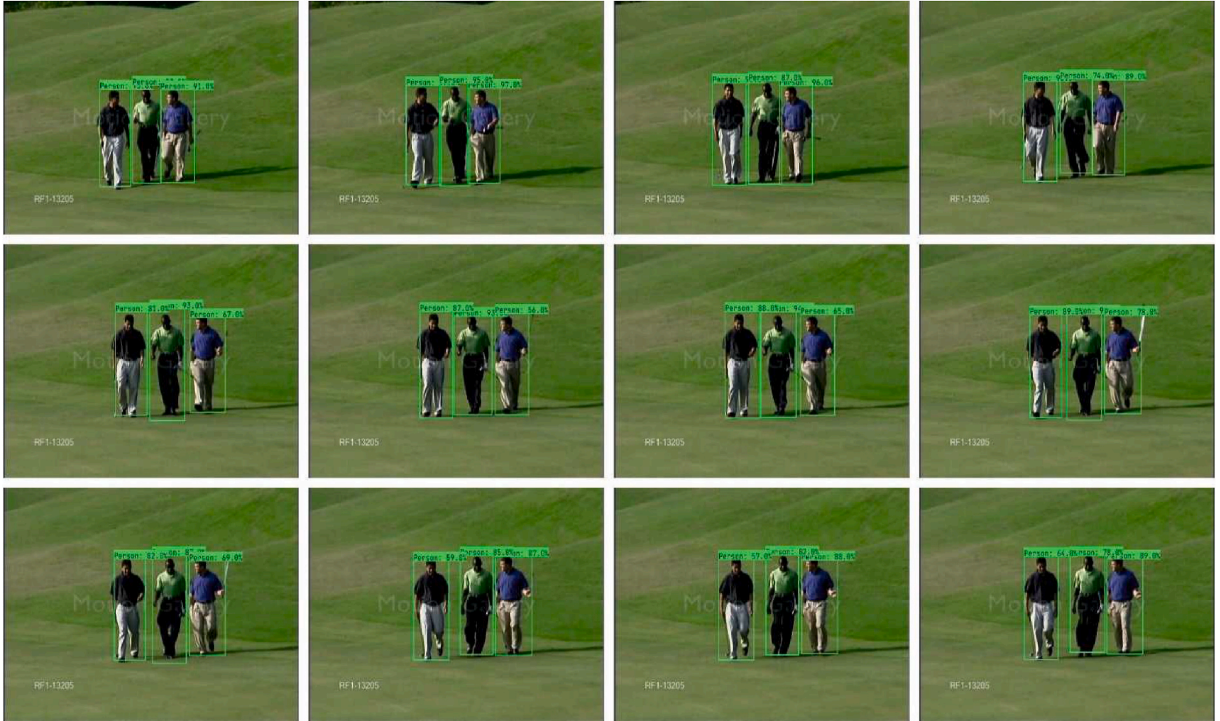


Fig. 10. Sample person detection on test video 2.

[20]. Fig. 9 shows the sample person detection and tracking results offered by the DLD-APDT model on test video 1. The figure implied that the DLD-APDT model has effectually detected and tracked the persons present in the input frames.

Fig. 10 depicts the sample person detection and tracking outcomes obtainable by the DLD-APDT approach on test video 2. The figure exposed that the DLD-APDT approach has effectively identified and

tracked the persons present from the input frames.

For demonstrating the better outcomes of the DLD-APDT model, a comparative analysis with existing models is made in Table 3 [21]. Fig. 11 provides a comparative study of the DLD-APDT method with recent DL algorithms interms of $prec_n$. The figure indicated that the EspiNetV2 model has shown ineffectual outcomes with lower $prec_n$ value of 62.79%. Meanwhile, the Faster R-CNN and YOLOv3 methodologies

Table 3
Comparative analysis of DLD-APDT approach with existing algorithms.

Measures	Precision	Recall
DLD-APDT Model	92.95	61.86
EspiNetV2 Model	62.79	59.85
Faster R-CNN Model	85.71	60.21
Yolov3 Model	85.65	60.17

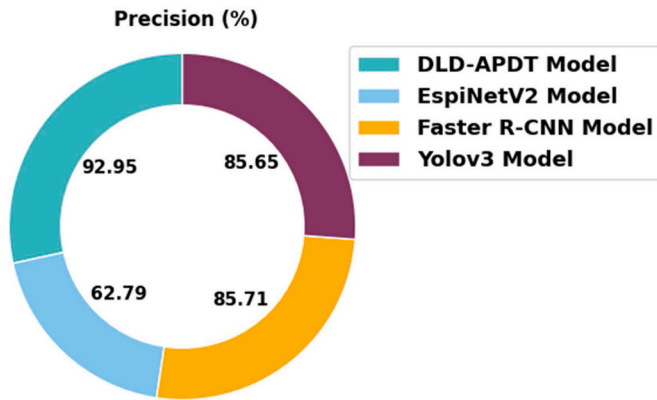


Fig. 11. analysis of DLD-APDT algorithm with existing methods.

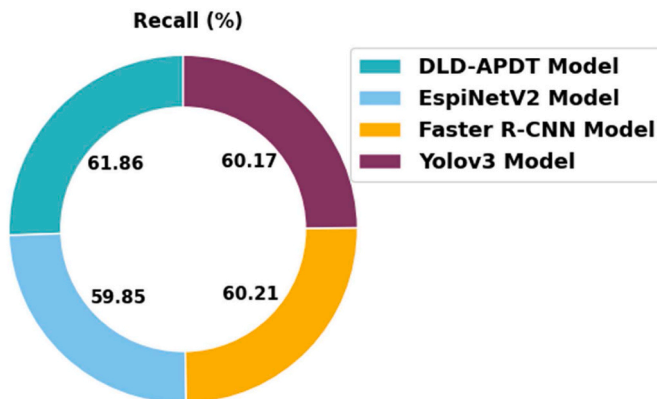


Fig. 12. analysis of DLD-APDT algorithm with existing methods.

have shown slightly enhanced detection performance with closer $prec_n$ values of 85.71% and 85.65% correspondingly. However, the presented DLD-APDT method has demonstrated superior performance with maximal $prec_n$ of 92.95%.

Fig. 12 demonstrates a comparison analysis of the DLD-APDT algorithm with recent DL models in terms of $recal_i$. The figure indicated that the EspiNetV2 system has shown ineffectual outcomes with lower $recal_i$ value of 59.85%. Also, the Faster R-CNN and YOLOv3 models have exposed somewhat maximal detection performance with closer $recal_i$ values of 60.21% and 60.17% correspondingly. Eventually, the presented DLD-APDT algorithm has outperformed superior performance with higher $recal_i$ of 61.86%. These results and discussion assured that the DLD-APDT model has demonstrated better people detection and tracking performance.

4. Conclusion

In this article, a new DLD-APDT model was enhanced for effectual person detection and tracking on surveillance videos. The proposed DLD-APDT model initially performs frame conversion process where the

input video is converted into a set of frames. Followed by, the proposed DLD-APDT model utilized EfficientDet model to detect persons and track them. Finally, the RMSProp optimizer is applied to optimally choose the hyperparameters related to the EfficientNet model which helps in accomplishing enhanced performance. The experimental validation of the DLD-APDT model takes place using two training datasets namely PascalVOC and PenFudan datasets. The simulation results reported the enhanced detection and tracking performance of the DLD-APDT technique over current approaches. Hence, the DLD-APDT model can be utilized as an effectual person detector in real time environment. In future, metaheuristics based hyperparameter optimizers can be applied to enhance the detection efficiency of the DLD-APDT model.

CRedit authorship contribution statement

S. Sivachandiran: Manuscript Updation, Grammar correction. K. Jagan Mohan: Conceptualization, Methodology. G. Mohammed Nazer: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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